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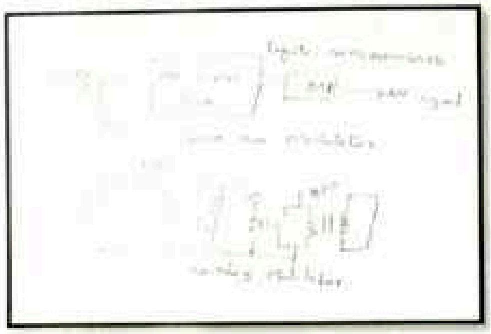
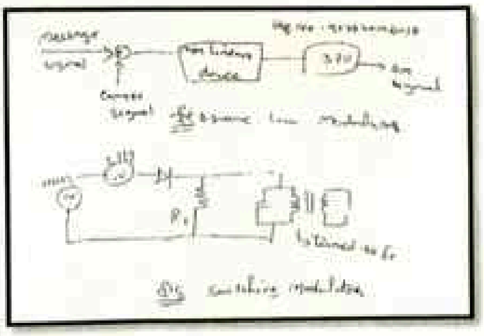
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

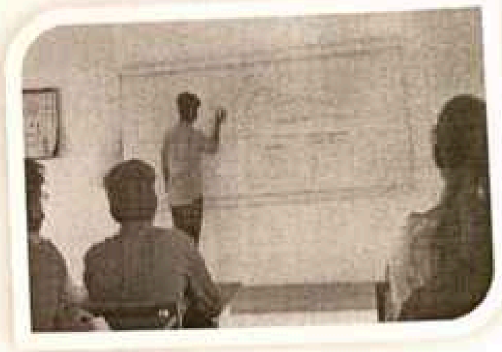
Department of Electronics and Communication Engineering
Academic Year: 2021- 2022 (Even Semester)

INNOVATIVE TEACHING METHOD

Degree, Semester& Branch: IV Semester B.E. ECE A
Course Code & Title: EC8491 Communication Theory
Name of the Faculty member: Mrs.G.Gnana Priya

Sl.No.	Topic(s)	Activity	Reference(s)
UNIT I - AMPLITUDE MODULATION			
1.	Amplitude Modulation	Role Play	J.G.Proakis, M.Salehi, "Fundamentals of Communication Systems", Pearson Education 2014
 			
2.	AM Generation – Square law and Switching modulator	Sketch noting	Simon Haykin, "Communication Systems", 4th Edition, Wiley, 2014
 			

3.	Revision-Unit I	Mind map	https://nptel.ac.in/courses/117105143



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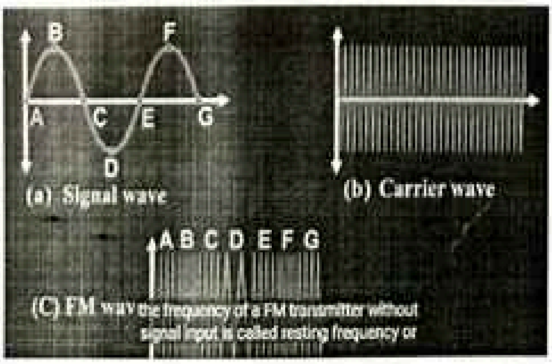
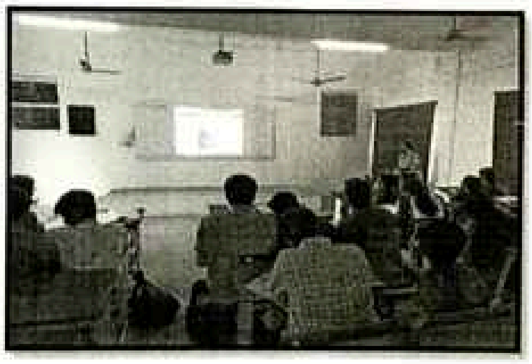
Sl.No.	Topic(s)	Activity	Reference(s)
UNIT II - ANGLE MODULATION			
1.	Narrow Band FM-Modulation index, Spectra, Power relations and Transmission Bandwidth	Zero Minute speech	J.G.Proakis, M.Salehi, "Fundamentals of Communication Systems", Pearson Education 2014


Sl.No.	Topic(s)	Activity	Reference(s)
2.	FM modulation –Direct method, Indirect Method	Animated video	Simon Haykin, "Communication Systems", 4th Edition, Wiley, 2014





3.	Revision-Unit II	Visible Quiz	https://nptel.ac.in/courses/117102059

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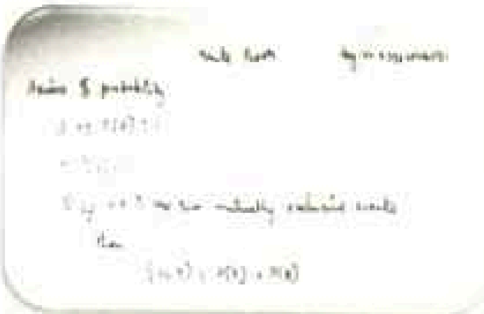
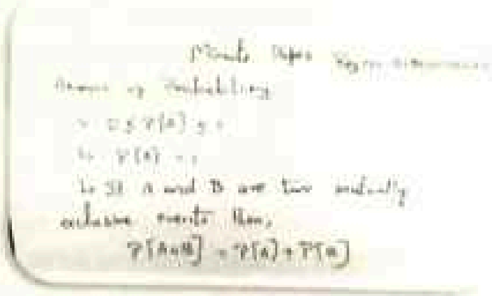
Degree, Semester& Branch: IV Semester B.E. ECE A
Course Code & Title: EC8491 Commuation Theory
Name of the Faculty member: Mrs.G.Gnana Priya

Sl.No.	Topic(s)	Activity	Reference(s)
	UNIT III - RANDOM PROCESS		
1.	Introduction to Probability models, Random Variables and Random Process	One Minute Paper	J.G.Proakis, M.Salehi, "Fundamentals of Communication Systems", Pearson Education 2014

Sl.No.	Topic(s)	Activity	Reference(s)
2.	Correlation and its properties	Think-Pair-Share	Simon Haykin, "Communication Systems", 4th Edition, Wiley, 2014


3.	Revision-Unit III	Quiz Dominos	https://nptel.ac.in/courses/117105085

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Innovative Practices Description for Unit-II

Degree, Semester& Branch: IV Semester B.E. ECE A

Course Code & Title: EC8491 Communication Theory

Name of the Faculty member: Mrs.G.Gnana Priya

Name of the Topic: Unit II-Revision

Name of the Innovative Practice: Visible Quiz

Description of Visible Quiz:

Students in groups discuss the appropriate response to quiz questions, typically multiple choices (A, B, C or D) or True (T) False (F). Each team has a set of large cards imprinted with one of the four letters. The cards also have a unique color. One person from each team displays the team's answer allowing the instructor to determine how well the students understood the question. Visible Quiz card have the advantage of allowing teachers to identify immediately the groups giving incorrect answers.

Goals (Learning Outcomes):

- The students will be able to classify the angle modulation schemes.
- The students will be able to explain the various generation and demodulation of angle modulation schemes
- Perform FM to PM conversion and vice-versa

Use of appropriate method:

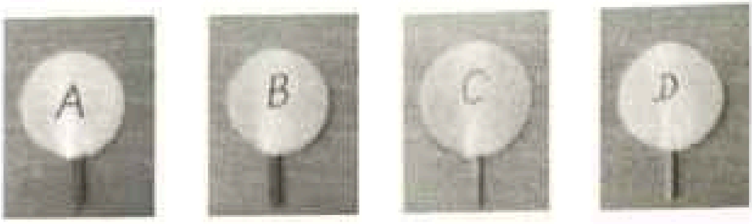
Justification for choosing the Activity:

The basics of frequency modulation are needed for understanding the concepts in the higher semester. So this activity is chosen to know the understanding level of the students. It enables the student to express their answers with clear ideas by raising the correct option.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

- The activity has been planned for 10 minutes with 10 questions from important topics of the unit.
- The entire class strength was divided into groups of 4 in a team (Class strength-49. Last team alone grouped with 5 members).
- Each member of the team was given with a card containing the options A,B,C,D.
- After displaying the questions the students were asked to raise the correct options in discussion with their team mates.

- The students were interested to answer the questions by showing the option cards.
- After that the correct answers were discussed in the class.



Sample Images:

Significance of Results: (Assessment of Effectiveness/Success of the Activity):

The students could be able to answer with the choices provided and differentiate the topics and clarify their doubts among them in this activity. The effectiveness of the activity will be measured in the Internal Assessment Test performance.

Reflective Critique:

Challenges:

The students took more time to discuss and answer each question. So the time duration to conduct the activity may be increased to conduct in a better way.

Benefits:

- This activity helped to easily recap the fundament concepts related to analog modulation.
- As this activity is a team-based activity, the collaborative learning skills of the students got increased.
- The students got a clear idea about the concept in the classroom itself.

References

1. J.G.Proakis, M.Salehi, "Fundamentals of Communication Systems", Pearson Education 2014.
2. https://nptel.ac.in/courses/117102059

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Planning document for Flipped Classroom (FC) activity

Unit / Topic: Unit IV / Noise Sources and its types
Course Outcome: CO4
Topic Learning Outcome: TLO 7

Objective:
To explain the various noise sources and its effect on communication systems.

Outcome:
After completion of this flipped classroom activity, the students will be able to
- Summarize the various noise sources in communication systems
- Analyze the effect of noise on communication systems.

Justification on selected topic for FC activity:

- The details of various noise sources are provided in Unit 4. The selected topic comprises of various noise sources and its types. Since the students are exposed to the basics of noise in communication systems, I decide to conduct a flipped class on '**Noise Sources and its types**'.
- Only a few types of noises from various sources were mentioned in their prescribed text book, despite the fact that there are many types of noise in communication systems. Students have the opportunity to search and explore more information related to specific types of noise if this flipped class is used in addition to the learning materials will be provided.
- The students can manage themselves and they will feel more comfortable to learn through video lectures and self-study reading materials.
- It is very much important topic because noise and its effect plays a major role in communication systems while transmitting and receiving the signal.

- The collaborative learning or peer tutoring can be easily done by the students for the simple theory concepts rather than for the complicated problematic topics. Hence this topic has been chosen for doing FC activity.

Learning materials (Video lectures and Book chapters)

Sl. No	Material Type	Details of the material
1	NPTEL Video	**Title:** Noise performance analysis in communication systems **Course name:** Communication Engineering **Professor:** Dr.Surendra Prasad, Department of Electrical Engineering, IIT Delhi **Time duration to be watched:** 35 minutes
2	Lecture Video	**Title:** Noise in Analog Communication Systems-A Conceptual Video Lecture **Time duration to be watched:** Full video (20.02 minutes) **Web link:** https://www.youtube.com/watch?v=gESgxUz8Bds
3	Book	**Title:** Communication Systems **Author:** Simon Haykin **Chapter Number:** 1 **Page Numbers:** 58-64
4	Book	**Title:** Fundamentals of Communication Systems **Author:** John G.Prokais **Chapter Numbers:** 6 **Page Number:** 255-262

The learning material consist of details of various noise sources such as

- Predictable & Unpredictable noise
- Correlated & Uncorrelated noise
- Atmospheric noise & Extra-terrestrial noise
- Solar noise & Cosmic noise
- Man-made noise
- Shot noise
- Partition noise
- Flicker noise & Transit time noise
- Thermal noise
- White noise

Details about how to use the learning resources:

- The students will be informed about the flipped class activity atleast 10 days before the implementation of the activity.

- The learning materials listed above will be used for Flipped Class activity. It will be intimated to the students through "Announcement section" of the course website Canvas LMS.
- The detailed email will be sent to the students regarding the best usage of the learning materials. The students will be informed to prepare a presentation for 10 minutes containing the
  - ✓ Introduction
  - ✓ Noise sources
  - ✓ Technical Information for the given topic
  - ✓ Impact of noise in communication systems
- With the following script, a short audio clip explaining how to use the learning materials will be created and sent to the students via WhatsApp.
  - ➢ Refer the LMS announcement section or email about the learning materials.
  - ➢ Two video lectures (Sl. No. 1-2) are listed. Kindly listen to at least any one of the video lectures. Both are equally good and cover the entire topic supposed to learn.
  - ➢ The animated video is excellent and it clearly explains "Types of noise in communication systems". So, all are instructed to watch the video for getting better clarity in the concept.
  - ➢ You can refer any one of the text book for remembering the basic concepts and to recollect the noise sources and its types.
  - ➢ Kindly prepare the PPT with points mentioned in Canvas LMS.

Strategy for Team formation

- The grouping of students will also be done well in advance and each group is allotted with one topic of their choice among the 10 noise sources listed. (10 days before implementation)
- In my class, I have 49 students. So, I will form a group of 5 students.
- I prefer heterogeneous team because it has advantages like natural support for peer coaching, easier classroom management, and better assistance to the low achievers. Hence ten heterogeneous group will be formed with students having different level of learning capabilities.
- Based on the performance of the students on previous class room activities, internal assessment tests, online quiz and assignment, they will be categorized into three. They are
  - ✓ High achievers
  - ✓ Average learners and
  - ✓ Low achievers

- Each group will have four students with one High achiever, two average learners and one low achiever.
- The team formation will be done well before the flipped class activity and it is informed to them well in advance. Hence the students can do out of the class activities i.e., group study together.

Plan of Activity for flipped classroom:

- The handout materials explaining the "Flipped Classroom" will be given to the students well in advance. The day before the learning materials are given to the students. I will explain about the Flipped Class activity and give the video link explaining the FC activity (https://www.youtube.com/watch?v=BCIxikOq73Q).

Sl. No.	Activity planned	Days	Time duration/day
1.	Out class activity • Watch the video • Read the material	Six days	30 minutes
2.	Out class activity • PPT preparation	Four days	05-15 minutes
3.	Out class activity • Discussions and clarification of doubts	Anytime within 10 days time for discussion within the group mates and any clarification regarding the flipped class	-
4	In class activity • Presentation • Question & Answering	Each group	10 minutes 3 minutes
	• Oral Feedback(At the end of the activity)	Each group	1 minute

Out – Class activity plan:

- The students are allowed to watch the videos and read the materials on their own pace for six days with minimum of 30 minutes a day.
- In Canvas a separate discussion thread will be created and it will be used for the clearing the doubts of the students either by other students or by the faculty.
- An announcement will be posted once in two days about the activity as reminder

In - class activity plan:

- The students have to prepare a presentation for 10 minutes with the concept learnt and 3 minutes for Question & answering and tell about the contribution of each member.

- Two hours will be used for conducting the Flipped Class in class activity. In which each group has to present their topic chosen and others will also listen to the presentation.
- The various noise sources are explained to the whole class and the hand-outs prepared on the topic will be shared within the group members.
- The main objective of the flipped class activity is each group will learn one noise source and the presentation helps the other members also to know about all the other type of noise sources.
- Thus, 10 types of noise sources will be completed in 2 periods through this activity.
- After completion of presentation, both oral and written feedbacks will be collected about the flipped class from the students.
- Each group will be assessed through the content of presentation, communication skills, Q& A, contribution by each member as shown below.
  - ✓ Content of presentation (20 Marks)
  - ✓ Communication skills (10 Marks)
  - ✓ Q & A(10 Marks)
  - ✓ Contribution (10 Marks)
- The marks will be considered for final internal mark calculation.

Difficulties may be faced in implementation and steps to overcome them:

- **No Preparation by the few students:** Some students may not read the book chapters well and may not watched the video lectures. In that case, they will not be able to participate in the flipped class activity well.
  - I will tell the students that few of the questions in final exam will be covered from these portions.
  - Enough time will be given to the students to watch the videos and study the materials.
  - Repeated reminders will be sent to the students through e-mail and LMS.
- **Non participation of some students in In-class activities:** Some students may not involve / participate in the activities planned. It may be avoided by
  - Encouraging the active students with small rewards like claps or candies.
  - Including the assessment mark for final internal mark calculation which makes each student to participate actively in the flipped class activity.
- **Time management:** I may not be able to complete the flipped class activity in the scheduled time.
  - The Time management can be done by using timer clock for 10 minutes duration.
- **Classroom control:** The students may create a noise while doing the activity.
  - Active participation of all the students will not make much noise. This can be avoided by repeatedly telling the students to reduce the noise level in the class room.

- If the activity is not planned and executed well, then the class will go out of control. This can be avoided by proper planning and execution.

Student's feedback:

The student's feedback will be collected at the end of the flipped class activity. It will be collected in two modes.

- **Oral feedback**
- **Written Feedback**

Oral Feedback:

The oral feedback will be collected at the end of the presentation by the students about

- Experience in doing this flipped class
- Contribution by each member of the group.
- What was the learning outcome of such activity?
- What is the difference you felt between traditional teaching and flipped class?

Written Feedback:

A written feedback will be collected with the following questionnaire at the end of completing the activity.

Students Feedback: Flipped Class Room Activity

1. How do you rate the activity for learning the concept of various noise sources?

   ☐ Excellent ☐ Good ☐ Satisfactory ☐ Poor

2. Video lectures and Pre-reading materials were available before the Flipped Classroom (FC) activity.

   ☐ Excellent ☐ Good ☐ Satisfactory ☐ Poor

3. Adequate time was provided to spend on watching the videos and reading the materials.

   ☐ Excellent ☐ Good ☐ Satisfactory ☐ Poor

4. Videos and materials are relevant to the FC activity.

   ☐ Excellent ☐ Good ☐ Satisfactory ☐ Poor

5. FC activity has improved my understanding of the key concepts.

   ☐ Excellent ☐ Good ☐ Satisfactory ☐ Poor

6. FC activity gave me an opportunity to communicate with my team members (students).

☐ Excellent ☐ Good ☐ Satisfactory ☐ Poor

7. Teacher was able to engage me in the FC activity.

☐ Excellent ☐ Good ☐ Satisfactory ☐ Poor

8. Teacher was able to add more clarity on the key concepts during FC activity.

☐ Excellent ☐ Good ☐ Satisfactory ☐ Poor

9. Teacher was able to control the classroom during FC activity.

☐ Excellent ☐ Good ☐ Satisfactory ☐ Poor

Share your experience in doing this FC activity

What improvements would you recommend to improve learning in the Flipped Classroom?

Name and Signature of the Student

References:

1. https://uwaterloo.ca/centre-for-teaching-excellence/teaching-resources/teaching-tips/planning-courses-and-assignments/course-design/course-design-planning-flipped-class
2. https://uwaterloo.ca/centre-for-teaching-excellence/teaching-resources/teaching-tips/lecturing-and-presenting/delivery/class-activities-and-assessment-flipped-classroom
3. https://uwaterloo.ca/centre-for-teaching-excellence/teaching-resources/teaching-tips/lecturing-and-presenting/delivery/online-activities-and-assessment-flipped-classroom
4. https://www.collegestar.org/modules/flipped-classroom
5. https://er.educause.edu/articles/2017/9/myths-and-facts-about-flipped-learning?utm_source=Informz&utm_medium=Email&utm_campaign=ER#_zsEDIKe1_zlG9oE4

Signature of the Faculty member

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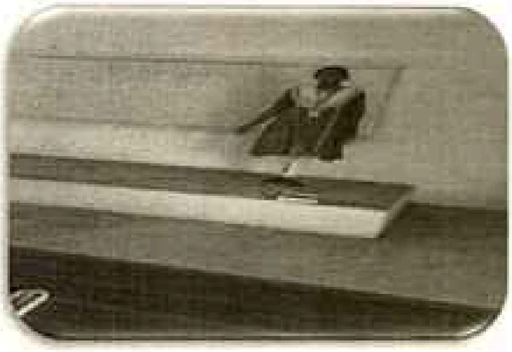
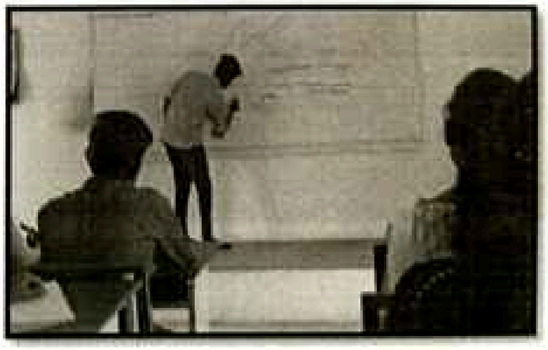
Name of the Faculty member: Mrs.G.Gnana Priya

Sl.No.	Topic(s)	Activity	Reference(s)
	UNIT IV - NOISE CHARACTERIZATION		
1.	Representation of Narrow band noise –In-phase and quadrature	Just a Minute	J.G.Proakis, M.Salehi, "Fundamentals of Communication Systems", Pearson Education 2014

Sl.No.	Topic(s)	Activity	Reference(s)
2.	Noise performance analysis in FM systems	Pro-Con Grid	Simon Haykin, "Communication Systems", 4th Edition, Wiley, 2014

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Sl.No.	Topic(s)	Activity	Reference(s)
UNIT V - SAMPLING & QUANTIZATION			
1.	Introduction to sampling & quantization	Quescussion	J.G.Proakis, M.Salehi, "Fundamentals of Communication Systems", Pearson Education 2014


Sl.No.	Topic(s)	Activity	Reference(s)
2.	Low pass sampling	Online Simulation	https://ocw.mit.edu/courses/6-003-signals-and-systems-fall-2011/resources/lecture-21-sampling/


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